
File: DIB_inte

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File: MCU.kicad

File: ADIB_PV

File: Voltage_

--

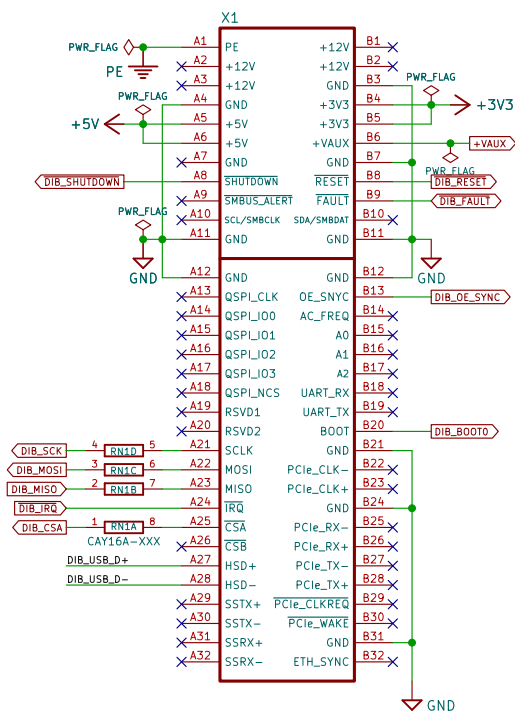
File: Voltage_

File: Digital_

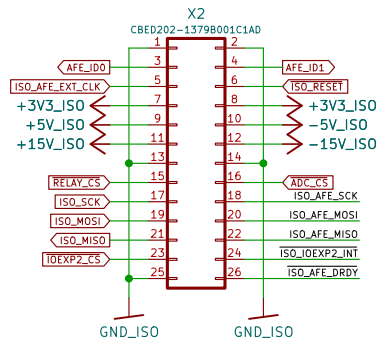


Id: 1/7

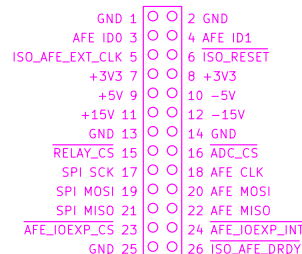
DIB2 connector



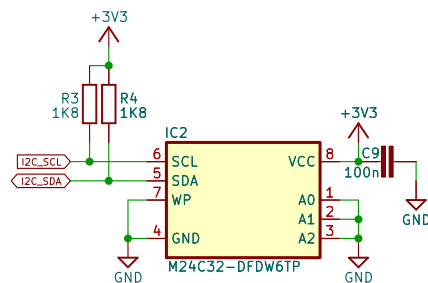
Isolated AFE connector



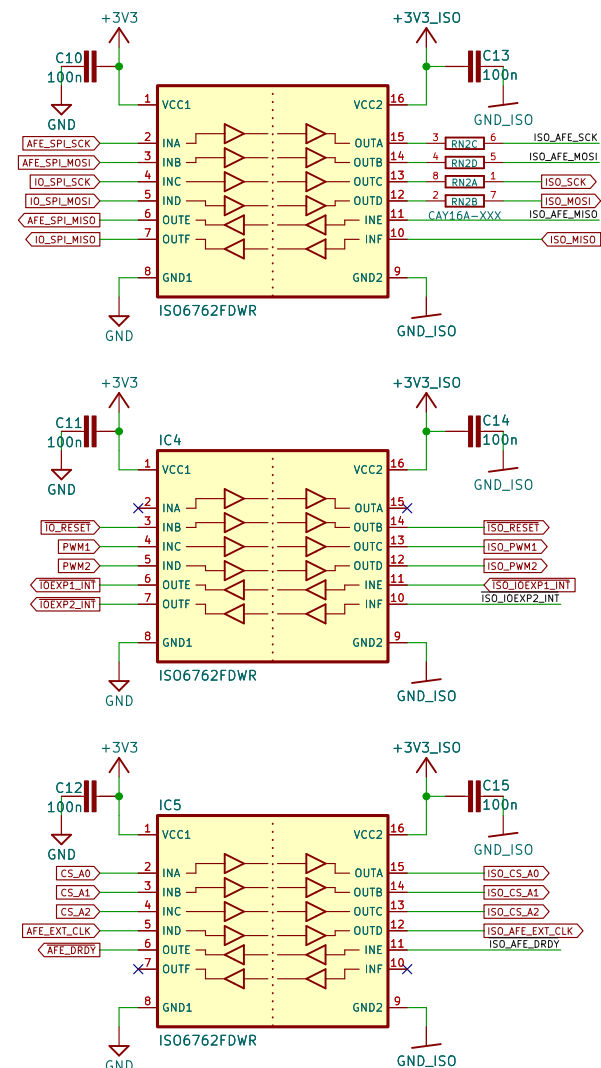
26-pin AFE connector



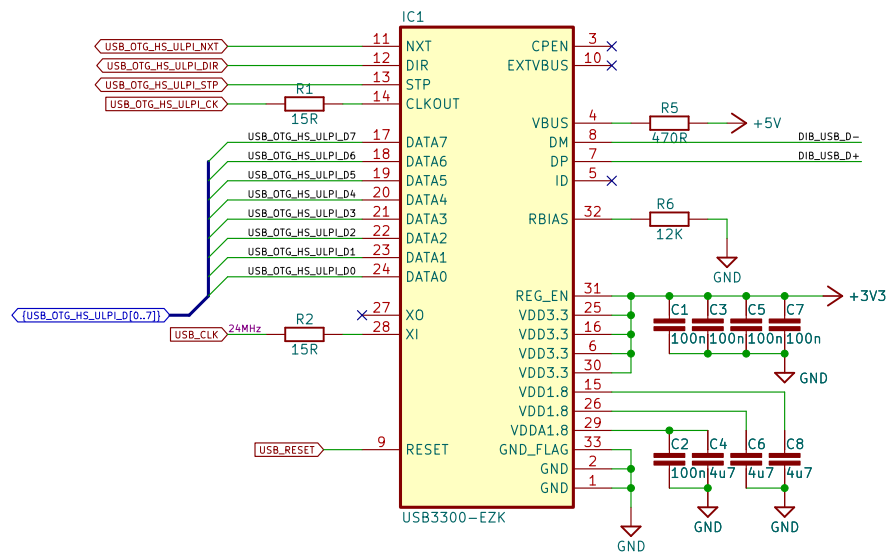
I2C Module ID EEPROM



Bus Isolator



USB HS PHY



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Sheet: /DIB2 interface, USB PHY/

File: DIB_interface.kicad_sch

Title: EEZ DIB MI0168+ module

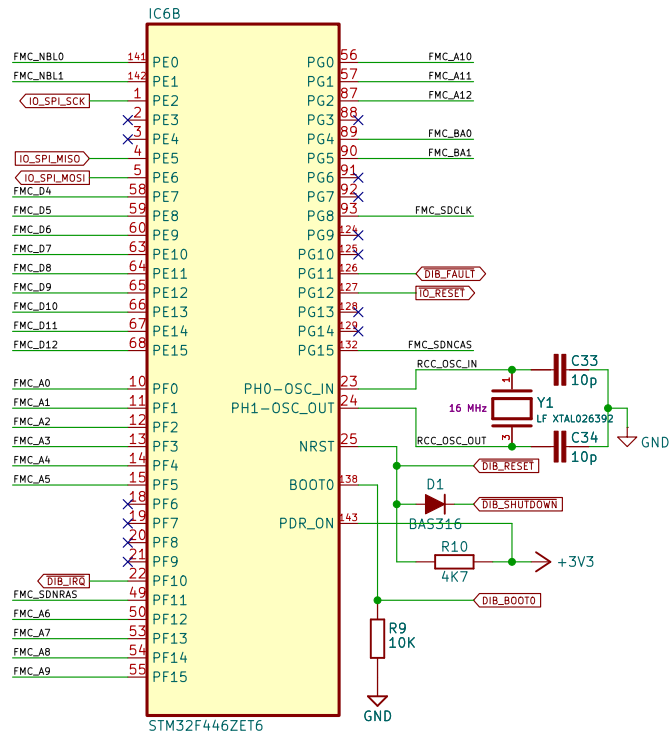
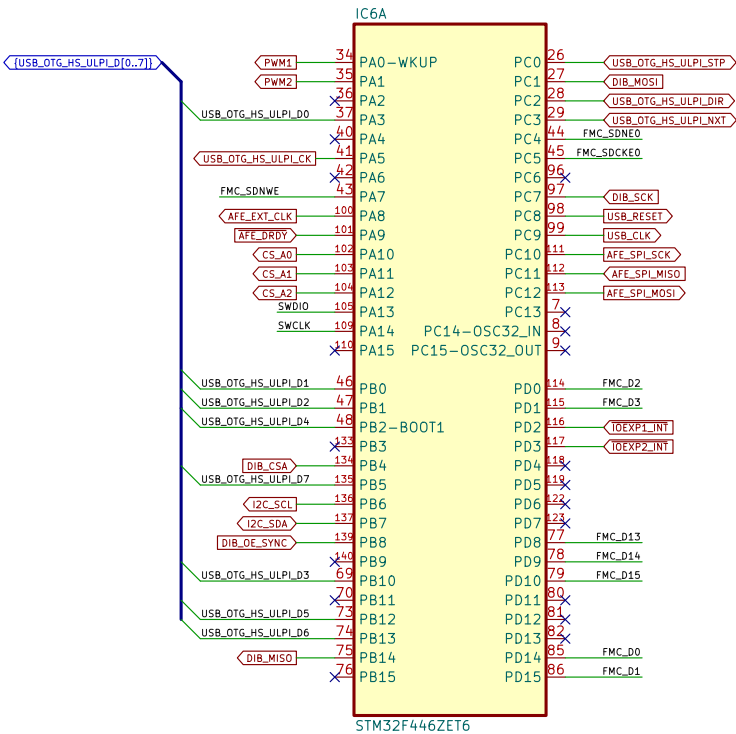
Size: A4 Date: 2026-07-21

KiCad E.D.A. 10.0.4

Rev: r1B1

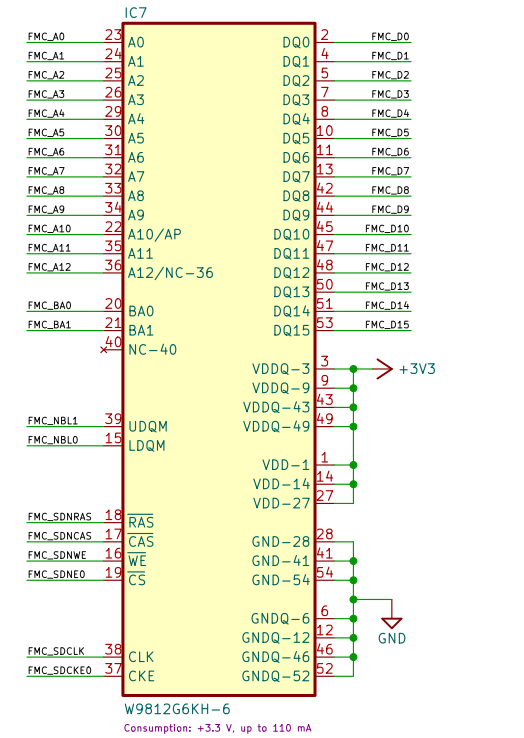
Id: 2/7

32-bit MCU



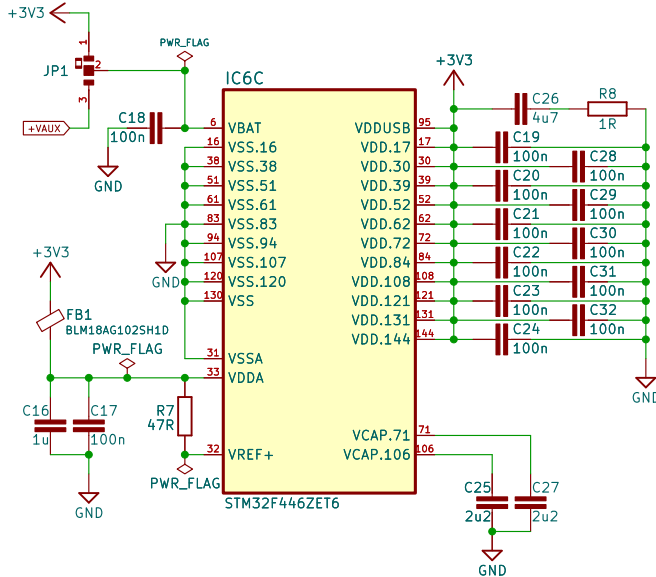
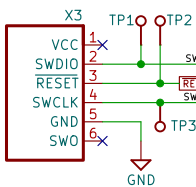
SDRAM

IMPORTANT: For PCB layout
consult ST AN4661 section 8.4.2



PCB pads for
Tag-Connect TC2030-CTX

SWO 6 10 15 GND
SWCLK 4 0 0 3 nRST
SWDIO 2 0 0 1 N.C.



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Sheet: /MCU/

File: MCU.kicad_sch

Title: EEZ DIB MIO168+ module

Size: A4

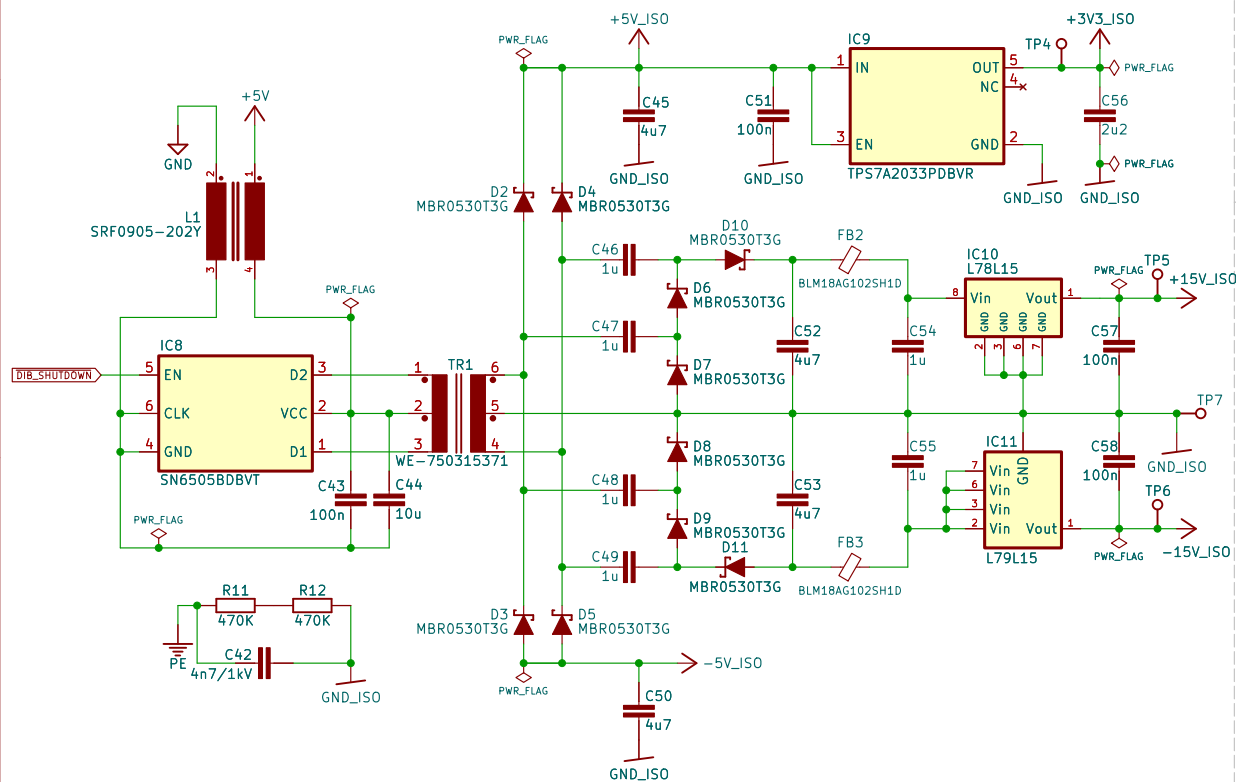
Date: 2026-07-21

KiCad E.D.A. 10.0.4

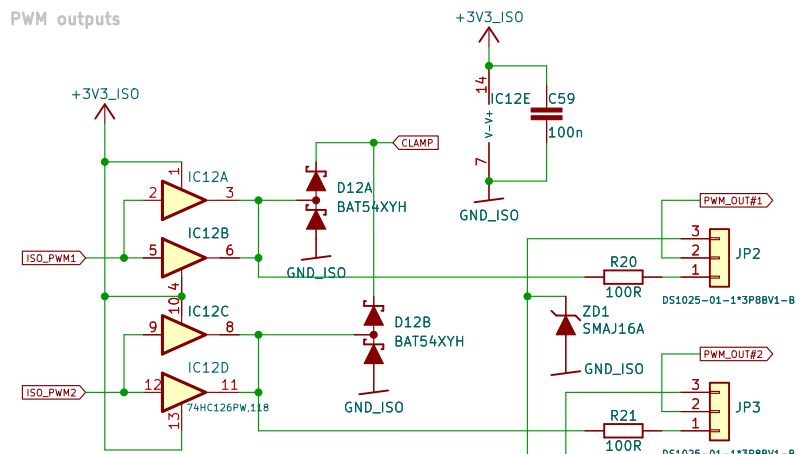
Rev: r1B1

Id: 3/7

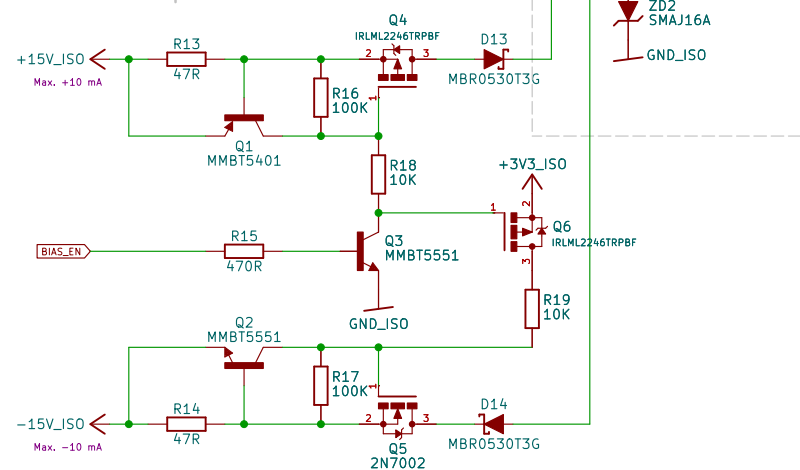
Isolated +/-5 V, +/-12 V power supply, +3.3 V LDO



PWM outputs



Sensor bias outputs



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Sheet: /ADIB, PWM, AFE connectors/

File: ADIB_PWM.kicad_sch

Title: EEZ DIB MIO168+ module

Size: A4

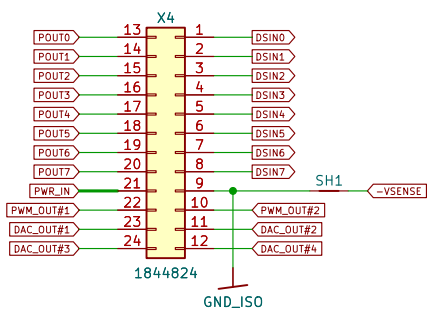
Date: 2026-07-21

KiCad E.D.A. 10.0.4

Rev: r1B1

Id: 4/7

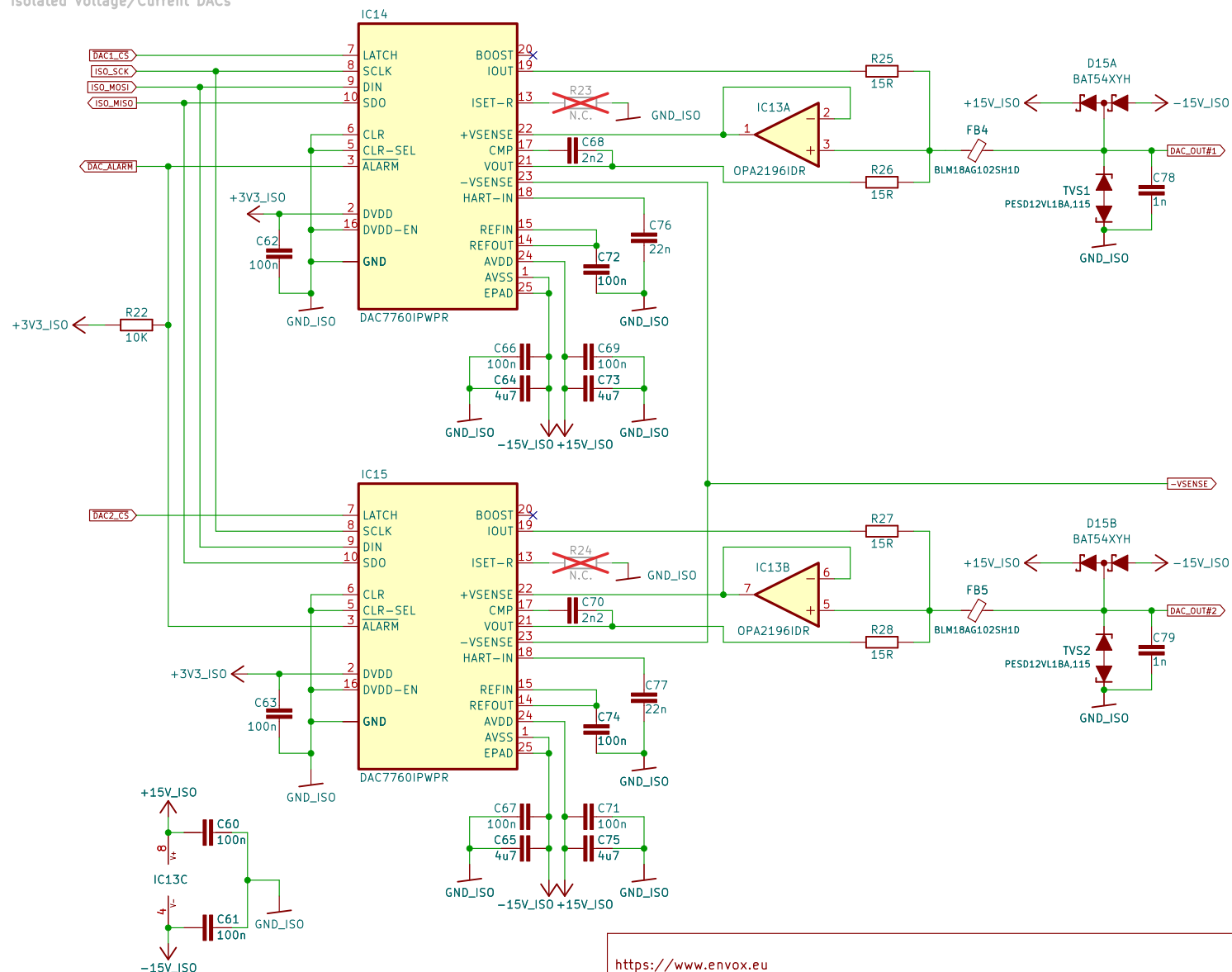
Isolated Front panel I/O connector



24-pin I/O connector

DOU0 13	1	DIN0
DOU1 14	2	DIN1
DOU2 15	3	DIN2
DOU3 16	4	DIN3
DOU4 17	5	DIN4
DOU5 18	6	DIN5
DOU6 19	7	DIN6
DOU7 20	8	DIN7/MISO
PWR IN 21	9	GND
PWM1/SCK 22	10	PWM2/MOSI
DAC #1 23	11	DAC #2
DAC #3 24	12	DAC #4

Isolated Voltage/Current DACs



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Sheet: /Voltage_current DACs/
File: Voltage_current_DACs.kicad_sch

Title: EEZ DIB MIO168+ module

Size: A4 Date: 2026-07-21
KiCad E.D.A. 10.0.4

Rev: r1B1
Id: 5/7

[illegible]

The image contains three circuit diagrams for the isolated SPI CS decoder, 14-bit I/O expander, and 8-ch digital power outputs.

Isolated SPI CS decoder: This diagram shows the SN74LVC138ADBDR (IC16) connected to the SPI CS decoder. The inputs are ISO_CS_A0 (pin 1), ISO_CS_A1 (pin 2), ISO_CS_A2 (pin 3), and ISO_RESET (pin 4). The outputs are RELAY_CS (pin 15), ADC_CS (pin 14), DAC1_CS (pin 13), DAC2_CS (pin 12), DAC34_CS (pin 11), DOUT_CS (pin 10), IOEXP1_CS (pin 9), and IOEXP2_CS (pin 7). The VCC (pin 16) is connected to +3V3_ISO, and GND (pin 8) is connected to GND_ISO. A 100nF capacitor (C80) is connected between VCC and GND.

Isolated 14-bit I/O Expander: This diagram shows the PCAL9714HNP (IC19) connected to the 14-bit I/O expander. The inputs are IOEXP1_CS (pin 16), ISO_MISO (pin 17), ISO_SCK (pin 19), ISO_MOSI (pin 20), ISO_RESET (pin 24), and ISO_IOEXP1_IN (pin 22). The outputs are DIN0 (pin 1), DIN1 (pin 2), DIN2 (pin 3), DIN3 (pin 4), DIN4 (pin 5), DIN5 (pin 6), DIN6 (pin 7), DIN7 (pin 8), SLOW_DIN (pin 10), LEVEL_DIN (pin 11), BIAS_EN (pin 12), AFE_ID0 (pin 13), AFE_ID1 (pin 14), and DAC_ALARM (pin 15). The VDD (pin 21) is connected to +3V3_ISO, and GND (pin 25) is connected to GND_ISO. A 100nF capacitor (C85) is connected between VDD and GND.

Isolated 8-ch digital power outputs: This diagram shows the TLE75008ESDXUMA1 (IC21) connected to the 8-ch digital power outputs. The inputs are ISO_SCK (pin 2), ISO_MOSI (pin 3), ISO_MISO (pin 4), and ISO_RESET (pin 21). The outputs are POUT0 (pin 6), POUT1 (pin 19), POUT2 (pin 7), POUT3 (pin 18), POUT4 (pin 9), POUT5 (pin 16), POUT6 (pin 12), and POUT7 (pin 13). The VDD (pin 24) is connected to +3V3_ISO, and GND (pin 25) is connected to GND_ISO. A 100nF capacitor (C88) is connected between VDD and GND. A PWR_FLAG (pin 20) is connected to PWR_IN (pin 22) through a 10uF capacitor (C91) and a Zener diode (ZD3, SMAJ24A/TR13).

The image contains three circuit diagrams for the isolated SPI CS decoder, 14-bit I/O expander, and 8-ch digital power outputs.

Isolated SPI CS decoder: This diagram shows the SN74LVC138ADBDR (IC16) connected to the SPI CS decoder. The inputs are ISO_CS_A0 (pin 1), ISO_CS_A1 (pin 2), ISO_CS_A2 (pin 3), and ISO_RESET (pin 4). The outputs are RELAY_CS (pin 15), ADC_CS (pin 14), DAC1_CS (pin 13), DAC2_CS (pin 12), DAC34_CS (pin 11), DOUT_CS (pin 10), IOEXP1_CS (pin 9), and IOEXP2_CS (pin 7). The VCC (pin 16) is connected to +3V3_ISO, and GND (pin 8) is connected to GND_ISO. A 100nF capacitor (C80) is connected between VCC and GND.

Isolated 14-bit I/O Expander: This diagram shows the PCAL9714HNP (IC19) connected to the 14-bit I/O expander. The inputs are IOEXP1_CS (pin 16), ISO_MISO (pin 17), ISO_SCK (pin 19), ISO_MOSI (pin 20), ISO_RESET (pin 24), and ISO_IOEXP1_IN (pin 22). The outputs are DIN0 (pin 1), DIN1 (pin 2), DIN2 (pin 3), DIN3 (pin 4), DIN4 (pin 5), DIN5 (pin 6), DIN6 (pin 7), DIN7 (pin 8), SLOW_DIN (pin 10), LEVEL_DIN (pin 11), BIAS_EN (pin 12), AFE_ID0 (pin 13), AFE_ID1 (pin 14), and DAC_ALARM (pin 15). The VDD (pin 21) is connected to +3V3_ISO, and GND (pin 25) is connected to GND_ISO. A 100nF capacitor (C85) is connected between VDD and GND.

Isolated 8-ch digital power outputs: This diagram shows the TLE75008ESDXUMA1 (IC21) connected to the 8-ch digital power outputs. The inputs are ISO_SCK (pin 2), ISO_MOSI (pin 3), ISO_MISO (pin 4), and ISO_RESET (pin 21). The outputs are POUT0 (pin 6), POUT1 (pin 19), POUT2 (pin 7), POUT3 (pin 18), POUT4 (pin 9), POUT5 (pin 16), POUT6 (pin 12), and POUT7 (pin 13). The VDD (pin 24) is connected to +3V3_ISO, and GND (pin 25) is connected to GND_ISO. A 100nF capacitor (C88) is connected between VDD and GND. A PWR_FLAG (pin 20) is connected to PWR_IN (pin 22) through a 10uF capacitor (C91) and a Zener diode (ZD3, SMAJ24A/TR13).

The image contains three circuit diagrams for the isolated SPI CS decoder, 14-bit I/O expander, and 8-ch digital power outputs.

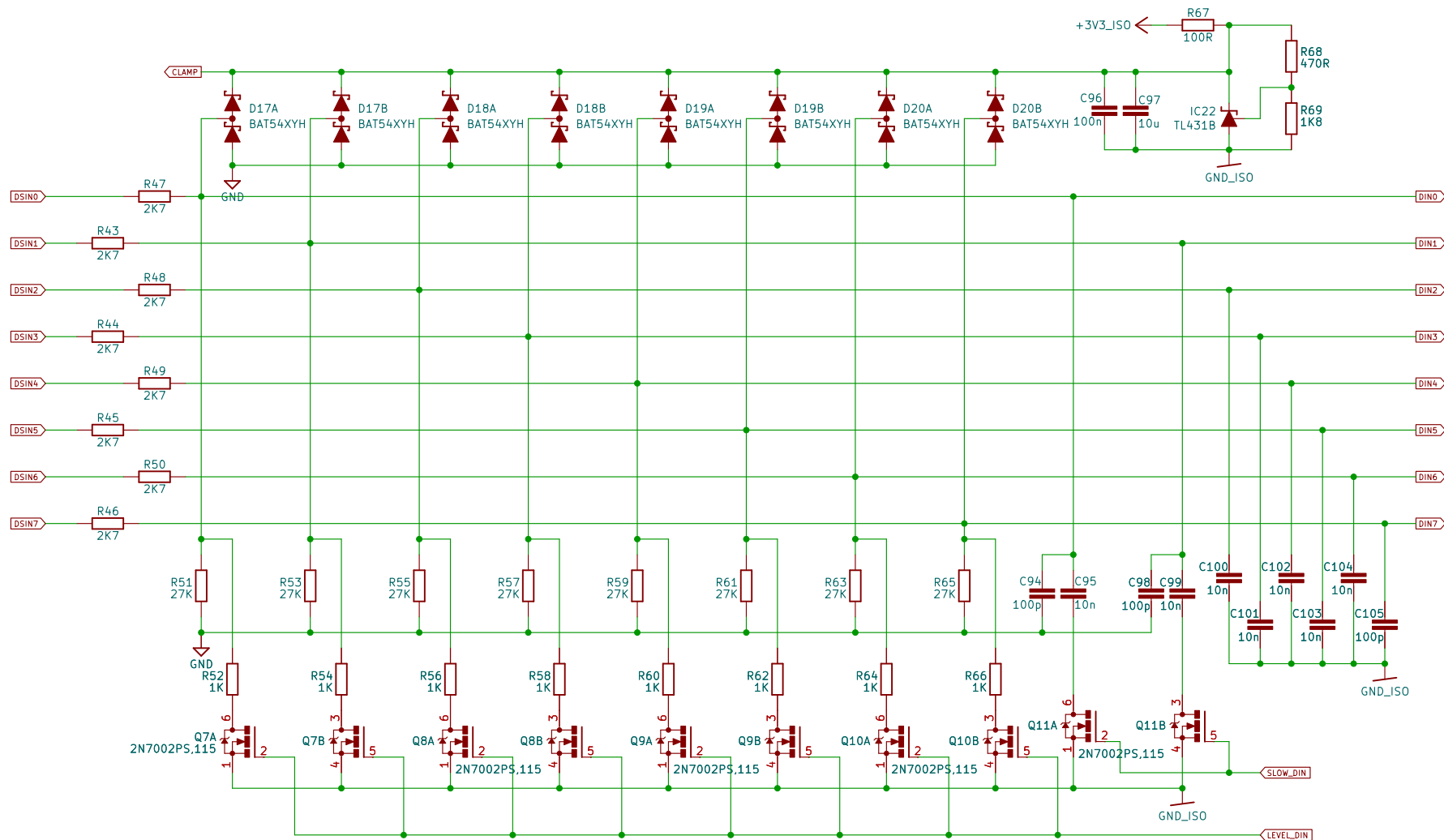
Isolated SPI CS decoder: This diagram shows the SN74LVC138ADBDR (IC16) connected to the SPI CS decoder. The inputs are ISO_CS_A0 (pin 1), ISO_CS_A1 (pin 2), ISO_CS_A2 (pin 3), and ISO_RESET (pin 4). The outputs are RELAY_CS (pin 15), ADC_CS (pin 14), DAC1_CS (pin 13), DAC2_CS (pin 12), DAC34_CS (pin 11), DOUT_CS (pin 10), IOEXP1_CS (pin 9), and IOEXP2_CS (pin 7). The VCC (pin 16) is connected to +3V3_ISO, and GND (pin 8) is connected to GND_ISO. A 100nF capacitor (C80) is connected between VCC and GND.

Isolated 14-bit I/O Expander: This diagram shows the PCAL9714HNP (IC19) connected to the 14-bit I/O expander. The inputs are IOEXP1_CS (pin 16), ISO_MISO (pin 17), ISO_SCK (pin 19), ISO_MOSI (pin 20), ISO_RESET (pin 24), and ISO_IOEXP1_IN (pin 22). The outputs are DIN0 (pin 1), DIN1 (pin 2), DIN2 (pin 3), DIN3 (pin 4), DIN4 (pin 5), DIN5 (pin 6), DIN6 (pin 7), DIN7 (pin 8), SLOW_DIN (pin 10), LEVEL_DIN (pin 11), BIAS_EN (pin 12), AFE_ID0 (pin 13), AFE_ID1 (pin 14), and DAC_ALARM (pin 15). The VDD (pin 21) is connected to +3V3_ISO, and GND (pin 25) is connected to GND_ISO. A 100nF capacitor (C85) is connected between VDD and GND.

Isolated 8-ch digital power outputs: This diagram shows the TLE75008ESDXUMA1 (IC21) connected to the 8-ch digital power outputs. The inputs are ISO_SCK (pin 2), ISO_MOSI (pin 3), ISO_MISO (pin 4), and ISO_RESET (pin 21). The outputs are POUT0 (pin 6), POUT1 (pin 19), POUT2 (pin 7), POUT3 (pin 18), POUT4 (pin 9), POUT5 (pin 16), POUT6 (pin 12), and POUT7 (pin 13). The VDD (pin 24) is connected to +3V3_ISO, and GND (pin 25) is connected to GND_ISO. A 100nF capacitor (C88) is connected between VDD and GND. A PWR_FLAG (pin 20) is connected to PWR_IN (pin 22) through a 10uF capacitor (C91) and a Zener diode (ZD3, SMAJ24A/TR13).

Id: 6/7

Digital inputs



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Sheet: /Digital inputs/
File: Digital_inputs.kicad_sch

Title: EEZ DIB MIO168+ module

Size: A4 Date: 2026-07-21
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Rev: r1B1
Id: 7/7